Lesson 10

1. What does `FILTER(Sales, Sales[Amount] > 1000)` return?

It returns a **table** containing only rows from the Sales table where `Amount > 1000`.

- Example:
  From the dataset:

```
SaleID | Amount
1      | 1200
3      | 1500
```

---

2. Write a measure High Sales that sums Amount where Amount > 1000 using FILTER.

```
High Sales =
CALCULATE(
    SUM(Sales[Amount]),
    FILTER(Sales, Sales[Amount] > 1000)
)
```

Result with dataset: `1200 + 1500 = 2700`.

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3. How does `ALLEXCEPT(Sales, Sales[Region])` differ from `ALL(Sales)`?

- `ALL(Sales)` removes **all filters** on the Sales table.
- `ALLEXCEPT(Sales, Sales[Region])` removes all filters **except Region**, so calculations remain grouped by Region.

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4. Use SWITCH to categorize Amount.

```
Amount Category =
SWITCH(
    TRUE(),
    Sales[Amount] > 1000, "High",
    Sales[Amount] >= 500 && Sales[Amount] <= 1000, "Medium",
    "Low"
)
```

Example: 1200 → High, 800 → Medium, 600 → Medium.

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5. What is the purpose of ALLSELECTED?

It returns all the rows visible after applying slicers, **but ignores filters inside individual visuals**. Used for percentage-of-total calculations **that respect slicers**.

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6. Write a measure Regional Sales % using ALLEXCEPT.

```
Regional Sales % =
DIVIDE(
    SUM(Sales[Amount]),
    CALCULATE(
        SUM(Sales[Amount]),
        ALLEXCEPT(Sales, Sales[Region])
    ),
    0
)
```

Shows the contribution of each sale to its region's total.

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7. Create a dynamic measure using SWITCH to toggle between SUM, AVERAGE, and COUNT of Amount.

```
Dynamic Measure =
SWITCH(
    SELECTEDVALUE(Metric[Value]),
    "SUM", SUM(Sales[Amount]),
    "AVERAGE", AVERAGE(Sales[Amount]),
    "COUNT", COUNT(Sales[Amount])
)
```

(Requires a slicer on a disconnected Metric table with values SUM, AVERAGE, COUNT).

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8. Use FILTER inside CALCULATE to exclude "Furniture" sales.

```
Sales Excluding Furniture =
CALCULATE(
    SUM(Sales[Amount]),
    FILTER(Products, Products[Category] <> "Furniture")
)
```

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9. Why might ALLSELECTED behave unexpectedly in a pivot table?

Because pivot tables (or matrix visuals) **apply additional context**.
If you use ALLSELECTED with no parameters, it respects slicers but **ignores row/column filters** inside the visual.

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10. Write a measure that calculates total sales and ignores filters from region.

```
Total Sales Ignore Region =
CALCULATE(
    SUM(Sales[Amount]),
    ALL(Sales[Region])
```

```
)
```

---

11. Optimize this measure:

❌ Old:

```
High Sales = CALCULATE(SUM(Sales[Amount]), FILTER(Sales, Sales[Amount] >
1000))
```

☑ Optimized:

```
High Sales =
CALCULATE(
    SUM(Sales[Amount]),
    Sales[Amount] > 1000
)
```

Using a Boolean filter is more efficient than wrapping in FILTER.

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12. Write a measure Top 2 Products using TOPN and FILTER.

```
Top 2 Products =
CALCULATE(
    SUM(Sales[Amount]),
    TOPN(2, VALUES(Sales[ProductID]), SUM(Sales[Amount]), DESC)
)
```

Returns sales only for the top 2 products.

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13. Use ALLSELECTED with no parameters.

```
Total Sales AllSelected =
CALCULATE(
    SUM(Sales[Amount]),
    ALLSELECTED()
)
```

Respects slicers but ignores visual-level filters.

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14. Debug: A SWITCH measure returns incorrect values in a matrix visual.

Possible issues:

- Missing `TRUE()` in SWITCH condition.
- Measure relies on row context (use SELECTEDVALUE).
- Visual adds multiple fields → causes ambiguity.

15. Simulate a "reset filters" button using ALL in a measure.

```
Sales Reset Filters =
CALCULATE(
    SUM(Sales[Amount]),
    ALL(Sales)
)
```

When added to a button/bookmark, it resets all filters.